May–June 2026 Study Plan

Nuclear #1 & Nuclear #2 — Missed-Chapter Catch-Up (with QCD & Weak-Interaction Review)

Owner: Parker • Budget: 10 hrs/week • 6 weeks • ~59 hrs total • Revised deadline: Sun Jun 14, 2026

Reading Sources

Short name	Full reference	Target chapters
Loveland	Loveland, Morrissey & Seaborg — Modern Nuclear Chemistry (Wiley, 2006)	8, 9, 10, 11, 12, 14, 18
Mann	Robert Mann — An Introduction to Particle Physics and the Standard Model (CRC, 2010)	3, 4, 5, 6, 9, 10, 15, 16, 17, 18, 19, 20, 21, 22, 23
LB	Lancaster & Blundell — Quantum Field Theory for the Gifted Amateur (Oxford, 2014)	7, 12, 13, 16, 17, 18, 19, 20

How the plan was sized

The original 21 target chapters (~400 pp) plus the QCD/weak add-on from Mann (Ch. 15–18 QCD + Ch. 19–23 weak/electroweak, ~130 pp) bring the total to ~30 chapters / ~530 pp. At 10 hrs/week the budget is roughly 2 hrs per chapter, split ~60% reading / ~40% problems. The six weeks ramp from concrete (nuclear decay & reactions) to abstract (symmetries, Feynman diagrams, QFT), then close the loop by staying in Mann for the QCD and weak / electroweak sectors.

Problem code: Loveland & Mann number their end-of-chapter problems 1, 2, 3… LB uses (7.1), (7.2)… When fewer than 5 are recommended, pick the listed ones first; if time remains, attempt the *bonus* in italics.

Month at a Glance

Week	Dates	Theme	Books / Chapters	Hours
Warm-up	Fri May 1 – Sun May 3	Review & setup	Skim old Nuc #1 notes; organize PDFs	~2
Week 1	Mon May 4 – Sun May 10	Nuclear decay & reactions	Loveland 8, 9, 10, 11	10
Week 2	Mon May 11 – Sun May 17	Nuclear applications	Loveland 12, 14, 18	10
Week 3	Mon May 18 – Sun May 24	Particle-physics foundations	Mann 3, 4, 5, 6, 9, 10	10
Week 4	Mon May 25 – Sun May 31	QFT: propagators / S-matrix / scattering	LB 7, 12, 13, 16, 17, 18, 19, 20	10
Week 5	Mon Jun 1 – Sun Jun 7	QCD review (Mann)	Mann 15, 16, 17, 18	9
Week 6	Mon Jun 8 – Sun Jun 14	Weak / electroweak review (Mann)	Mann 19, 20, 21, 22, 23	10

Suggested daily cadence: 2 hrs × 5 weekdays (or 2.5 hrs × 4 days). Keep one weekend day as buffer/review. Total = 10 hrs/week.

Week 1 • May 4 – May 10 • Nuclear Decay & Reactions

Goal: rebuild intuition for the three big decay modes and how nuclear reactions are parameterized. This is the densest page count of the month (~125 pp), so keep reading fast and force yourself to stop and do problems.

Loveland Ch. 8 — Beta (β) Decay

~21 pp • reading ~1.5 hr • problems ~1 hr

Problems: 1, 3, 5 *(bonus: 7)*

Focus: Neutrino hypothesis, Kurie plots, β-decay rate constant, parity non-conservation.

Loveland Ch. 9 — Gamma (γ)-Ray Decay

~27 pp • reading ~1.5 hr • problems ~1 hr

Problems: 1, 2, 4 *(bonus: 6)*

Focus: Multipolarity, internal conversion, Mössbauer effect.

Loveland Ch. 10 — Nuclear Reactions

~48 pp • reading ~2 hr • problems ~1 hr

Problems: 1, 3, 5

Focus: Cross sections, Rutherford & elastic scattering, compound-nucleus vs. direct reactions.

Loveland Ch. 11 — Fission

~30 pp • reading ~1.5 hr • problems ~0.5 hr

Problems: 2, 4 *(bonus: 6)*

Focus: Liquid-drop barrier, fission product mass/charge distributions, TKE.

End-of-week check-in (Sun May 10): can you sketch the spectral shape for β-decay, write down $\sigma(E)$ for Rutherford, and explain why fission fragments have an asymmetric mass distribution? If not, re-read the summaries before Monday.

Week 2 • May 11 – May 17 • Nuclear Applications

Goal: finish the Loveland block with the applied/experimental chapters. Ch. 14 and 18 are the longest of the month but largely descriptive — use them to lock in the vocabulary you'll be expected to know.

Loveland Ch. 12 — Nuclear Reactions in Nature: Nuclear Astrophysics

~32 pp • reading ~2 hr • problems ~1 hr

Problems: 1, 3, 5

Focus: Primordial nucleosynthesis, stellar burning chains, solar-neutrino problem.

Loveland Ch. 14 — Reactors & Accelerators

~44 pp • reading ~2 hr • problems ~1 hr

Problems: 1, 3, 5 *(bonus: 7)*

Focus: Light-water reactors, ion sources, cyclotron & synchrotron basics, beam transport.

Loveland Ch. 18 — Radiation Detectors

~40 pp • reading ~2.5 hr • problems ~1.5 hr

Problems: 1, 2, 4 *(bonus: 6)*

Focus: Gas ionization, semiconductor & scintillation detectors, Poisson statistics & upper limits.

End-of-week check-in (Sun May 17): you should now be able to pick the right detector for a given particle/energy, and explain the difference between a thermal reactor's fuel cycle and a fast reactor's. Take Sunday off if you're ahead — Week 3 gets abstract.

Week 3 • May 18 – May 24 • Particle-Physics Foundations (Mann)

Goal: jump into Nuc #2 territory. The Mann chapters are short but formalism-heavy (groups, Noether, C/P/T, scattering amplitudes). Stay in the problem sets — you can't fake fluency here.

Mann Ch. 3 — Symmetries

~17 pp • reading ~1 hr • problems ~0.75 hr

Problems: 1, 4, 5 *(bonus: 8)*

Focus: Groups, Lie algebras, SO(3), structure constants.

Mann Ch. 4 — Conservation Laws

~12 pp • reading ~0.75 hr • problems ~0.5 hr

Problems: 1, 2, 3

Focus: Action principle, Noether currents for translations/rotations.

Mann Ch. 5 — Particle Classification

~15 pp • reading ~1 hr • problems ~0.5 hr

Problems: 1, 2, 4

Focus: Spectroscopic notation, adding angular momenta, Clebsch-Gordan.

Mann Ch. 6 — Discrete Symmetries

~16 pp • reading ~1 hr • problems ~0.75 hr

Problems: 1, 3, 5 *(bonus: 7)*

Focus: P, T, C individually and the CPT theorem; positronium puzzle.

Mann Ch. 9 — Scattering

~16 pp • reading ~1 hr • problems ~0.75 hr

Problems: 1, 2, 4

Focus: Cross sections, 2-body decay rate, detailed balance.

Mann Ch. 10 — A Toy Theory

~20 pp • reading ~1 hr • problems ~0.75 hr

Problems: 1, 3, 5

Focus: First-pass Feynman rules in a scalar toy model — drill this; it's the entry point to Week 4.

End-of-week check-in (Sun May 24): can you derive Noether's theorem in one line, list the CPT assumptions, and sketch a tree-level Feynman diagram in Mann's toy theory? If the last one is shaky, re-read Ch. 10 before Monday — LB will assume you have it.

Week 4 • May 25 – May 31 • QFT: Propagators, S-matrix, Scattering (LB)

Goal: move into the densest material. LB chapters are short (~5–12 pp) but each one introduces a genuinely new machine. Budget 1 problem per chapter and do not skip the exercises — LB's exercises are unusually illuminating.

LB Ch. 7 — Examples of Lagrangians — how to write down a theory

~6 pp • reading ~0.5 hr • problems ~0.5 hr

Problems: (7.1), (7.2)

Focus: Massless/massive scalar fields, ϕ^4 theory, complex scalar.

LB Ch. 12 — Examples of Canonical Quantization

~8 pp • reading ~0.75 hr • problems ~0.5 hr

Problems: (12.1), (12.2)

Focus: Complex scalar field and its Noether current; non-relativistic limit.

LB Ch. 13 — Fields with Many Components & Massive Electromagnetism

~9 pp • reading ~0.75 hr • problems ~0.5 hr

Problems: (13.1), (13.2)

Focus: Internal symmetries, polarization/projection operators.

LB Ch. 16 — Propagators & Green's Functions

~9 pp • reading ~0.75 hr • problems ~0.5 hr

Problems: (16.1), (16.2)

Focus: Green's functions as inverse differential operators; the propagator picture.

LB Ch. 17 — Propagators & Fields

~10 pp • reading ~0.75 hr • problems ~0.5 hr

Problems: (17.1), (17.2)

Focus: Feynman propagator, Yukawa's force carriers.

LB Ch. 18 — The S-matrix

~10 pp • reading ~0.75 hr • problems ~0.5 hr

Problems: (18.1), (18.2)

Focus: Interaction picture, perturbation expansion, Wick's theorem.

LB Ch. 19 — Expanding the S-matrix: Feynman Diagrams

~12 pp • reading ~1 hr • problems ~0.75 hr

Problems: (19.1), (19.2), (19.3)

Focus: ϕ^4 theory diagrams, symmetry factors, p-space calculations.

LB Ch. 20 — Scattering Theory

~7 pp • reading ~0.5 hr • problems ~0.5 hr

Problems: (20.1), (20.2)

Focus: Yukawa $\psi^\dagger\psi\phi$ theory, transition matrix, cross-section.

End-of-week check-in (Sun May 31): self-test by writing, from memory, the Feynman rules for LB's Yukawa theory and deriving the tree-level cross-section. Once that's solid, roll straight into Week 5 — the QCD chapters build directly on the S-matrix / Feynman rules you just learned.

Week 5 • Jun 1 – Jun 7 • QCD Review (Mann)

Goal: close the loop on the strong interaction. Mann Ch. 15–18 walk you from the discovery of quarks and color through heavy flavors to the QCD Lagrangian and Yang-Mills appendix. Feynman-rules fluency from Week 4 is a prerequisite — if QCD diagrams are still confusing on Wednesday, re-read LB Ch. 19 before pushing on.

Mann Ch. 15 — From Nuclei to Quarks

~18 pp • reading ~1.25 hr • problems ~0.75 hr

Problems: 1, 3, 5

Focus: Range of the nuclear force, isospin, strangeness, flavor, color — the empirical staircase that forces $SU(3)_{color}$ *on us.*

Mann Ch. 16 — The Quark Model

~17 pp • reading ~1.25 hr • problems ~0.75 hr

Problems: 1, 2, 4 *(bonus: 6)*

Focus: Baryon & meson multiplets, mass relations, magnetic moments; why the quark model predicts what it does.

Mann Ch. 17 — Testing the Quark Model

~20 pp • reading ~1.25 hr • problems ~1 hr

Problems: 1, 2, 4

Focus: Vector-meson decay, Rosenbluth formula, deep-inelastic scattering, quark structure functions — the experimental case for point-like quarks.

Mann Ch. 18 — Heavy Quarks and QCD

~28 pp • reading ~1.75 hr • problems ~1 hr

Problems: 1, 3, 5 *(bonus: 7)*

Focus: Charm (J/ψ, OZI rule), bottom, the QCD Lagrangian, asymptotic freedom, and the Yang-Mills appendix (§18.5) — work the Feynman-rules section end-to-end; this is Mann's own field-theory treatment of QCD.

End-of-week check-in (Sun Jun 7): write down the QCD Lagrangian from memory (straight from Mann §18.4–18.5), identify every term, and draw the gluon self-interaction vertices; explain in one paragraph why asymptotic freedom is a consequence of the non-abelian gauge structure. If either is shaky, spend Sunday re-reading §18.5 before starting Week 6.

Week 6 • Jun 8 – Jun 14 • Weak & Electroweak Review (Mann)

Goal: finish the Standard-Model story entirely out of Mann. Ch. 19–23 walk you from Fermi's four-fermion theory to the full SU(2)×U(1) electroweak model with spontaneous symmetry breaking, including Mann's own appendix on electroweak Feynman rules (§23.4) — that appendix is the field-theoretic payoff.

Mann Ch. 19 — From Beta Decay to Weak Interactions

~22 pp • reading ~1.25 hr • problems ~0.75 hr

Problems: 1, 2, 4

Focus: Fermi's theory, V–A current, neutrino properties and helicity, kaon oscillation and CP.

Mann Ch. 20 — Charged Leptonic Weak Interactions

~15 pp • reading ~1 hr • problems ~0.75 hr

Problems: 1, 3, 5

Focus: Neutrino-electron scattering, muon decay, 3-body phase space. The appendix on ε-tensors and phase space is worth working through by hand.

Mann Ch. 21 — Charged Weak Interactions of Quarks & Leptons

~24 pp • reading ~1.25 hr • problems ~0.75 hr

Problems: 1, 2, 4 *(bonus: 6)*

Focus: Neutron decay, pion decay, quark–lepton vertices, CKM mixing. Keep a running glossary of the vertex factors.

Mann Ch. 22 — Electroweak Unification

~14 pp • reading ~1 hr • problems ~0.5 hr

Problems: 1, 3, 5

Focus: Neutral currents, ν-e and e^+e^- neutral scattering, the SU(2)×U(1) model, Weinberg angle.

Mann Ch. 23 — Electroweak Symmetry Breaking

~29 pp • reading ~1.75 hr • problems ~1 hr

Problems: 1, 3, 5 *(bonus: 7)*

Focus: Higgs mechanism, breaking SU(2), gauge-boson masses, and the Feynman-rules appendix for electroweak theory (§23.4) — do the appendix alongside the chapter; this is your field-theoretic treatment of the weak sector.

End-of-month check-in (Sun Jun 14): from memory, write the electroweak Lagrangian before and after symmetry breaking, identify the $W^{\pm}$, Z^0, photon, and Higgs, and derive $M_W/M_Z = \cos\theta_W$. If that works, you are caught up on the missed readings for both Nuclear #1 and Nuclear #2 **and** on the QCD / weak-interaction review.

Progress Tracker (print & tick off)

Week	Chapter	Read	Problems	Notes
1	Loveland 8 – Beta Decay	■	■	
1	Loveland 9 – Gamma-Ray Decay	■	■	
1	Loveland 10 – Nuclear Reactions	■	■	
1	Loveland 11 – Fission	■	■	
2	Loveland 12 – Nuclear Astrophysics	■	■	
2	Loveland 14 – Reactors & Accelerators	■	■	
2	Loveland 18 – Radiation Detectors	■	■	
3	Mann 3 – Symmetries	■	■	
3	Mann 4 – Conservation Laws	■	■	
3	Mann 5 – Particle Classification	■	■	
3	Mann 6 – Discrete Symmetries	■	■	
3	Mann 9 – Scattering	■	■	
3	Mann 10 – A Toy Theory	■	■	
4	LB 7 – Example Lagrangians	■	■	
4	LB 12 – Canonical Quantization Examples	■	■	
4	LB 13 – Many Components / Massive EM	■	■	
4	LB 16 – Propagators & Green's Functions	■	■	
4	LB 17 – Propagators & Fields	■	■	
4	LB 18 – The S-matrix	■	■	
4	LB 19 – Feynman Diagrams	■	■	
4	LB 20 – Scattering Theory	■	■	
5	Mann 15 – From Nuclei to Quarks	■	■	
5	Mann 16 – The Quark Model	■	■	
5	Mann 17 – Testing the Quark Model	■	■	
5	Mann 18 – Heavy Quarks & QCD (+ §18.5 Yang-Mills)	■	■	
6	Mann 19 – Beta Decay → Weak Interactions	■	■	
6	Mann 20 – Charged Leptonic Weak	■	■	
6	Mann 21 – Charged Weak: Quarks & Leptons	■	■	
6	Mann 22 – Electroweak Unification	■	■	
6	Mann 23 – Electroweak Symmetry Breaking (+ §23.4 Feynman rules)	■	■	

Slip plan: if you lose a week, drop Loveland Ch. 14 to skim-only and drop LB Ch. 13. The must-not-skip chapters are Loveland 8, 10, 11 and LB 17, 18, 19, 20 on the nuclear/QFT side, and Mann 15, 18, 22, 23 on the QCD / electroweak side — everything downstream assumes them. In Week 5, Mann §18.5 (Yang-Mills appendix) is the single most important section; in Week 6, Mann §23.4 (Feynman rules for electroweak theory) is the single most important section — do not skip either.